

SOLUTION TO MIDTERM QUESTION PAPER

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I(a). Let $A = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$ be the 2×2 matrix and a_i 's and b_j 's are independent Bernoulli(1/2) random variable for $i = 1, 2$ and $j = 1, 2$. A matrix has full rank iff determinant is non-zero.

$$\therefore \det(A) = a_1 b_2 - a_2 b_1.$$

We need to find that probability that the determinant is non-zero i.e., $P(\det(A) \neq 0) = 1 - P(\det(A) = 0)$.

Now, $\det(A) = 0$ iff $a_1 b_2 = a_2 b_1$.

$\therefore a_i$'s and b_j 's are independent Bernoulli(1/2) random variable, $a_1 b_2$ and $a_2 b_1$ can take only 0 or 1.

$$P(a_1 b_2 = 1) = P(a_1 = 1 \text{ and } b_2 = 1) = P(a_1 = 1)P(b_2 = 1) = \frac{1}{4}, \quad P(a_1 b_2 = 0) = 1 - P(a_1 b_2 = 1) = \frac{3}{4}.$$

Similarly,

$$P(a_2 b_1 = 1) = P(a_2 = 1)P(b_1 = 1) = \frac{1}{4}, \quad P(a_2 b_1 = 0) = \frac{3}{4}.$$

$$\begin{aligned} P(\det(A) = 0) &= P(a_1 b_2 = a_2 b_1) \\ &= P(a_1 b_2 = 0 \text{ and } a_2 b_1 = 0) + P(a_1 b_2 = 1 \text{ and } a_2 b_1 = 1) \\ &= P(a_1 b_2 = 0)P(a_2 b_1 = 0) + P(a_1 b_2 = 1)P(a_2 b_1 = 1) \\ &= \frac{5}{8}. \end{aligned}$$

Finally, we can write $p = P(\det(A) \neq 0) = 1 - P(\det(A) = 0) = \frac{3}{8}$.

I(b). Number of independent experiment conducted is 75, i.e. $n = 75$.

Y is the number of cases where the matrix has full rank and the probability is $p = \frac{3}{8}$.

$$\therefore Y \sim \text{Binomial}(75, \frac{3}{8}).$$

Now,

$$\begin{aligned} q = P(Y < 25) &= \sum_{k=0}^{24} {}^{75}C_k \left(\frac{3}{8}\right)^k \left(1 - \frac{3}{8}\right)^{75-k} \\ &\approx 0.5394, \text{ using the code } \text{sum(pbinom(seq(0,24),75,3/8))}. \end{aligned}$$

I(c). Approximate the distribution of Y using the CLT. We know, $Y \sim \text{Binomial}(75, \frac{3}{8})$.

$$E(Y) = np = 28.125$$

$$\begin{aligned} \text{Var}(Y) &= np(1-p) \approx 17.578 \\ \text{sd}(Y) &\approx 4.193. \end{aligned}$$

$\therefore$ By the CLT, Y is approximately normally distributed with mean 28.125 and sd 4.193.

Now, applying the continuity correction, we can write $P(Y < 25) = P(Y \leq 24.5)$.

Let $Z = \frac{Y-\mu}{\sigma} = \frac{24.5-28.125}{4.193} \approx -0.865$. $\therefore q = P(Y < 25) \approx P(Z < -0.864) \approx \Phi(-0.865) \approx 0.194$, using function `pnorm(-0.865,0,1)`. Φ represents the distribution function of normal distribution.

2(a). 4491.

2(b). Married (2815) using `length(which(dsub$MaritalStatus=="Married"))`.

2(c). Good (1549) using `length(which(dsub$HealthGen=="Good"))`.

2(d). (Married, Vgood) (1037) using `xtabs(~MaritalStatus+HealthGen,dsub)`.

3(a). Height is in centimeter and weight is in kilogram.

3(b). 4454.

3(c). 4451.

3(d). 4457.

4(a). $\text{Weight} = -34.95903 + 0.67662 \times \text{Height}$.

4(b). $\text{Weight} = -34.95903 + 0.67662 \times 182.9 = 88.79477$, 6 feet is equal to 182.88cm $\approx$ 182.9cm.

4(c). $\log(\text{Weight}) = 2.779660 + 0.009282 \times \text{Height}$.

4(d). $\text{Weight} = 88.0001$.